

Do improvements in ESG enhance returns? Exploring the ESG momentum strategy

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Abstract

This study investigates the ESG Momentum strategy using Sustainalytics ESG Ratings for British companies. It finds that shorter formation periods consistently generate reliable positive excess returns, particularly notable in a 3-month formation and 6-month holding strategy, delivering a monthly excess return of 0.63%. Conversely, 12-month ESG Momentum strategies yield negative returns even after accounting for growth and value factors. This phenomenon can be attributed to ESG Losers' extended period outperformance and slower market adjustments, contributing to favourable risk-adjusted returns for ESG Winners in shorter timeframes, establishing the viability of the ESG Momentum strategy. Notably, ESG Losers maintain positive monthly returns, suggesting positive stock performance despite ESG downgrades, rendering shorting ESG Losers ineffective in consistently enhancing risk-adjusted returns for ESG Momentum strategies. In shorter formation periods, ESG Momentum shows potential to outperform benchmarks, particularly due to ESG Winners' consistent outperformance, even as ESG Losers exhibit favourable short-term risk-adjusted returns. Integrating a price momentum tilt further improves the strategy's performance, enhancing returns by approximately 0.13% for the 3-month formation and 6-month holding strategy.

Client Specification

ESG will continue to hold a central focus for both investors and asset managers. Consequently, the growth in sustainability and ESG-related investment products is expected to gain significant momentum in the forthcoming years. Concurrently, research in the field of ESG and sustainability has been expanding, particularly within the realm of ESG integration in investment strategies and its impact on stock returns. Nevertheless, the empirical results from these analyses exhibit considerable divergence, and a consensus on sustainability remains an ongoing demand.

This paper employs corporate ESG ratings to investigate how enhancements in ESG ratings could yield favourable returns for investors seeking investment strategies capable of delivering competitive returns, while also aligning with their demand for sustainable practices—or at the very least, those that are striving to enhance their sustainability efforts. To achieve this, we studied the ESG Momentum strategy where we seek to understand stocks returns between companies that exhibit significant enhancements in ESG ratings and those that show comparatively marginal improvements. We conducted the study using publicly-listed British companies, categorizing them into groups based on changes in their ESG Ratings.

Our analysis underscores that shorter ESG Momentum periods consistently generate positive excess returns, particularly evident in a 3-month formation and 6-month holding strategy, yielding a monthly excess return of 0.63%. In contrast, 12-month ESG Momentum strategies display negative returns, even after accounting for growth and value factors. Additionally, this strategy demonstrates the potential to outperform benchmarks due to the sustained outperformance of ESG Winners, even as ESG Losers exhibit favourable short-term risk-adjusted returns.

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1. INTRODUCTION

The rise of ESG has introduced significant considerations and impact in shaping investment strategies. ESG integration has grown increasingly important, becoming a central focus for investment managers in the coming years. A 2022 report by PwC indicates that around 70% of investment managers plan to retrofit or adapt their products for ESG compliance. The report forecasts that ESG-oriented assets will experience faster growth compared to the entire Asset and Wealth Management market.

This study is prompted by the recognition of the growing significance of sustainable practices and the potential for improved non-financial risk management to enhance investment performance. The main objective of this paper is to analyse the influence of ESG considerations, particularly ESG ratings, on stock returns by investigating the ESG momentum strategy. This strategy builds upon the well-known Price Momentum strategy identified by Jegadeesh and Titman (1993), with the addition of corporate ESG Ratings that assess how companies manage environmental, social, and governance risks within their operations.

The purpose of analysing ESG momentum is to understand its approach to ESG integration and its potential to enhance financial performance, particularly through improved stock returns. The study aims to explore whether strong ESG risk management leads to better investment performance over time, showcasing the benefits of sustainable practices.

By comparing ESG 'Winners' and 'Losers,' we gain insights into the link between ESG improvements and investment performance. This analysis assesses the viability of the ESG momentum strategy in generating favourable risk-adjusted returns. The study evaluates ESG momentum across various formation and holding periods, comprehensively analysing the duration and sustainability of these effects.

This research aims to highlight ESG's importance in investment decisions and its impact on long-term financial returns. Understanding ESG momentum's influence on investment performance offers insights to investors, businesses, and stakeholders interested in responsible investment practices.

2. LITERATURE REVIEW

The resurgence of ESG has led to a variety of research surrounding ESG factors and its impacts on investment performance and portfolio returns. A prominent argument for ESG considerations is that it increases operational efficiency within the firm in the long run. Govindan et al. (2020) studied supply chain sustainability practices and found a positive relationship between these practices and firm performance. The study even shows the relationship has been growing stronger over time and has found it particularly stronger for firms in manufacturing relative to those in services.

Hong-Yi Chen et al. (2022) conducted an analysis of the ESG Momentum strategy in Taiwan and Japan stock markets but observed it to be a viable strategy only in the former country. The

strategy would last up to three years after portfolio formation, despite average ESG scores being lower than comparable peers in Japan.

Another recent work by Sverner et al. (2022) uses stocks that composes the S&P500 and estimated that companies with relatively worst ESG ratings exhibits an ‘ESG Risk Premium’ of 0.20% - 0.29% relative to companies with the best ratings. The paper also conducted an analysis of an ESG Momentum factor and saw that stock price appreciation were more evident in rating upgrades, with the strategy yielding a return premium of 0.23% - 0.35%.

3. METHODOLOGY

3.1. Data Collection

This study uses publicly listed British companies from November 2009 to Dec 2019, comprising a sample of 353 companies. Financial data are obtained from Refinitiv Compustat database while corporate ESG ratings, including monthly environmental, social, governance and weighted ratings, are extracted from Morningstar Sustainalytics. The databases are matched using the International Securities Identification Number (ISIN) code for each listed company.

Ranked on a scale of 0 to 100, we used Sustainalytics ESG ratings to gauge companies' ESG readiness and performance. Higher ratings indicate stronger ESG risk management practices compared to lower-rated counterparts. This study concentrates on two principal categories: ESG Winners and ESG Losers. ESG Winners denote companies that have undergone significant ESG rating upgrades, whereas ESG Losers pertain to those with noteworthy ESG rating downgrades.

3.2. Data Preview

Table 1: Summary Statistics of Key Variables

	<i>Mean</i>	<i>S.D.</i>	<i>Median</i>	<i>Min</i>	<i>Max</i>
<i>ESG</i>	61.56	9.04	61.58	36.00	89.00
<i>Environmental</i>	59.32	13.17	58.90	30.00	96.00
<i>Social</i>	61.04	10.44	61.00	29.00	92.83
<i>Governance</i>	66.00	9.83	66.00	34.29	97.00
<i>Monthly Returns</i>	0.55%	12.02%	0.41%	-89.43%	977.0%

Based on Table 1, our studied sample have a monthly average and median ESG rating of 61.56 and 61.58 respectively. The relatively small mean-median spread across the rating categories indicates a balanced distribution of environmental, social, and governance ratings. This combines with the limited impact of outliers, provides a solid foundation for conducting a meaningful analysis of the ESG momentum strategy. Amongst ESG components, we see that average for Environmental ratings are the lowest while also having the highest standard

deviation, suggesting that environmental risks display a relatively high level of variability compared to other components of ESG.

3.3. Portfolio Construction

Our investment strategy is based upon the momentum of ESG Ratings for each company for three formation period: – 3-, 6- and 12-months. For each formation period, we sorted each company into quintiles based on their respective absolute variations in ESG Ratings. Due to infrequent occurrence of alterations in ESG ratings, we decided to only include companies that have experienced movements in rating so we can better categorize companies into ESG Momentum groups. We will then invest in the top 20% i.e. the quintile with the highest improvement in ESG ratings (ESG Winners) and short the bottom 20% (ESG Losers). Similar to Jegadeesh and Titman (1993) price momentum strategy, we will adopt a zero-investment, equal weighting strategy which entails taking differences in average returns between both quintiles. As for frequency of portfolio rebalancing, our analysis will involve adopting 3-, 6-, and 12-month holding periods, which will see us analysing investment performance of 9 ESG momentum portfolios. The notations for these portfolios will follow the format of:

- ***F[formation period] - H[holding period]***

For instance, F3-H12 denotes a portfolio formed over 3 months and rebalanced every 12 months.

Statistical significance of our returns will be determined using a two-tailed t-test and the Fama-French 3-factor models which accounts for growth and value factors.

4. STRATEGY ANALYSIS & FINDINGS

4.1. ESG Momentum Strategies across Portfolios

Table 2: Returns of ESG Momentum Strategies

<i>Formation Period</i>	<i>Holding Period</i>	<i>P5 (Winners)</i>	<i>P4</i>	<i>P3</i>	<i>P2</i>	<i>P1 (Losers)</i>	<i>Winners - Losers</i>
3	3	0.66%	0.25%	-0.09%	0.53%	0.30%	0.35% **
3	6	0.97%	0.23%	0.02%	0.46%	0.33%	0.64% ***
3	12	1.01%	0.57%	0.87%	0.51%	0.94%	0.06%
6	3	0.63%	0.60%	0.12%	0.37%	0.74%	-0.11%
6	6	0.74%	0.39%	-0.22%	0.47%	0.55%	0.19%
6	12	0.82%	0.47%	0.25%	0.38%	0.57%	0.26% **
12	3	0.36%	0.39%	0.70%	0.78%	0.47%	-0.10%
12	6	0.30%	0.53%	0.60%	0.97%	0.54%	-0.24% **
12	12	0.37%	0.63%	0.87%	0.20%	0.50%	-0.13% **

Notations of *, **, and *** represents statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 2 shows the average monthly returns for our ESG Momentum strategy, tested for significance using a two-tailed t- test. Amongst the nine portfolios, 5 generated statistically significant returns. Within the 3-month formation portfolios, the -H3 and -H6 portfolio generated statistically significant returns of 0.35% and 0.64% respectively whereas only one portfolio within the 6-month formation, the -H12 portfolio, delivered significant positive returns.

Shifting our focus to the F12 groups, an intriguing observation emerges – all portfolios recorded negative returns. This intriguing outcome suggests a dynamic where ESG Losers exhibit a tendency to outperform their ESG-winning counterparts within the context of 12-month formation periods. This nuanced insight encourages a deeper exploration of the interplay between ESG factors and longer-term investment strategies.

Table 3: Fama-French 3 Factor Model for ESG Momentum Strategies

<i>Formation Period</i>	<i>Holding Period</i>	<i>Intercept</i>	<i>Mkt-rf</i>	<i>SMB</i>	<i>HML</i>	<i>Adj. R-squared</i>
3	3	0.0035 ** (2.16)	-0.0421 (-1.1)	0.0630 (0.62)	0.0687 (0.87)	-0.0098
3	6	0.0063 *** (3.99)	-0.0152 (-0.41)	-0.0367 (-0.37)	0.0585 (0.76)	-0.0201
3	12	0.0002 (0.09)	0.0451 (0.96)	-0.2992 ** (-2.4)	0.0057 (0.06)	0.0401
6	3	-0.0016 (-0.78)	0.0378 (0.78)	-0.0211 (-0.16)	0.0362 (0.36)	-0.0151
6	6	0.0016 (1.35)	-0.0084 (-0.31)	-0.0202 (-0.28)	0.0135 (0.24)	-0.0259
6	12	0.0023 ** (2.6)	-0.0354 ** (-1.69)	0.0185 (0.33)	-0.0073 (-0.17)	0.0116
12	3	-0.0014 (-0.84)	-0.0058 (-0.14)	0.0301 (0.29)	0.0309 (0.38)	-0.0268
12	6	-0.0027 ** (-2.38)	-0.0065 (-0.23)	0.0260 (0.37)	0.0658 (1.2)	-0.0128
12	12	-0.0017 ** (-1.99)	-0.0141 (-0.67)	0.0168 (0.32)	0.0032 (0.08)	-0.0225

The corresponding t-statistics are enclosed in parentheses. Notations of *, **, and *** represents statistical significance at the 10%, 5%, and 1% levels, respectively.

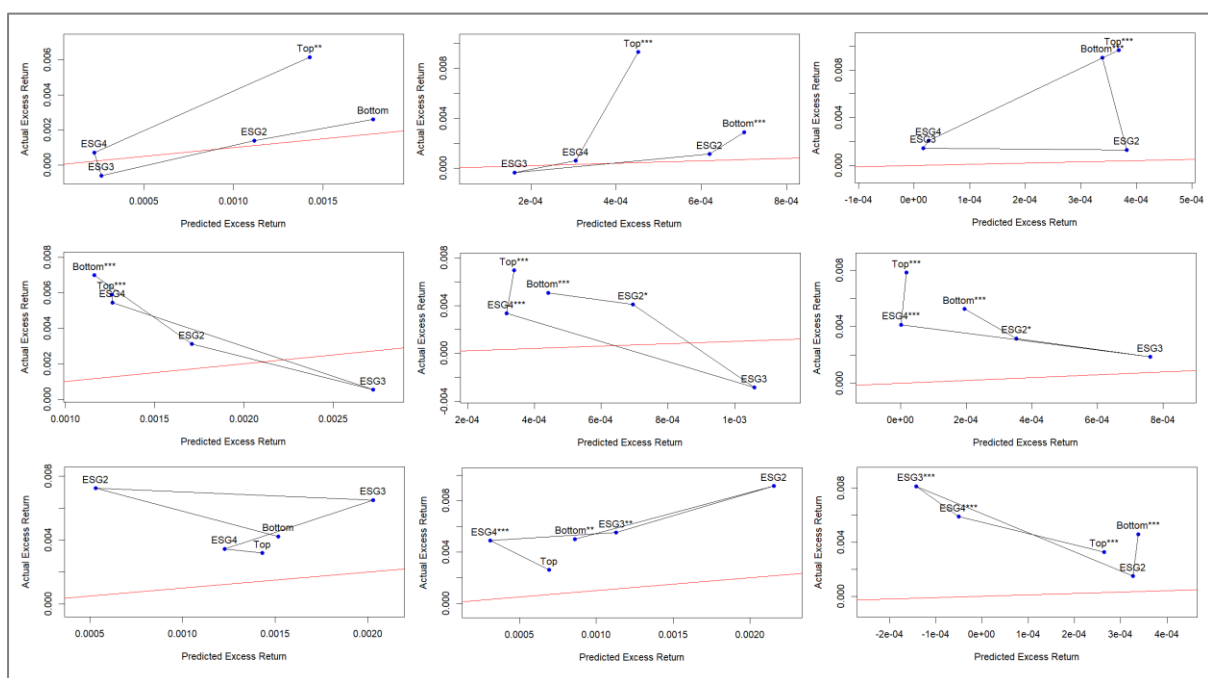
Table 3 shows the Fama-French 3-factor regression results for our ESG Momentum strategy. Stripping out value and growth effects, our Fama-French 3-factor (FF3) model indicates that three out of our nine portfolios had a statistically significant and positive alpha. The F3-H3 portfolio generated an alpha of 0.35%, significant at the 5% level, while the highest alpha is generated by our F3-H6 portfolio with 0.63% and is statistically significant at the 1% level.

The F6-H12 portfolio also had a statistically significant alpha at the 5% level, although at a more subtle 0.23%. Again, we see that our F12 portfolios all generates negative alphas.

This finding provides additional support that, particularly for short formation and holding periods, there exist a significant difference in returns between companies that experienced improvements in ESG risk management compared to those that are worse off, and this difference cannot be attributed to value or growth factors.

Another crucial finding is that the strategy returns does not benefit from shorting ESG Losers as these groups still deliver positive returns for all our studied portfolios. Despite that, for shorter formation periods, we could infer that ESG winners would outperform ESG Losers while the opposite is true when we observe longer formation periods.

Figure 1: Plot of Actual versus FF3 Predicted Returns



Notations of *, **, and *** represents statistically significant alphas at the 10%, 5%, and 1% levels, respectively. Red line represents 45° line. Rows: F3, F6, F12. Columns: -H3, -H6, -H12

Interestingly, we find that in most of our portfolios, both ESG Winners and Losers tend to generate significantly positive alphas as we observe in six out of nine of our ESG Momentum portfolios in Figure 1. Therefore, we conclude that a strategy that shorts the ESG Losers can only generate positive and meaningful returns only if there is a significant difference between the alphas of Winners and Losers, as is the case for shorter formation periods, specifically the F3-H3 and F3-H6 portfolio. As for the F12 portfolios, it is often the case that ESG Losers had generated relatively higher alphas compared to the Winners, resulting in negative returns for our ESG Momentum strategy.

Table 4: Excess Returns, Volatility and Sharpe Ratio of ESG Momentum Strategy

Formation	Holding	Excess Returns			Volatility			Sharpe Ratio		
		W	L	W - L	W	L	W - L	W	L	W - L
3	3	0.62%	0.26%	0.35% **	2.29%	2.02%	1.68%	0.27	0.13	0.21
3	6	0.93%	0.29%	0.64% ***	1.85%	1.29%	1.63%	0.50	0.22	0.40
3	12	0.96%	0.90%	0.06%	1.29%	2.23 %	2.14%	0.75	0.41	0.03
6	3	0.59%	0.70%	-0.11%	2.03%	2.31%	2.16%	0.29	0.30	-0.05
6	6	0.69%	0.51%	0.19%	1.37%	1.35%	1.20%	0.51	0.37	0.16
6	12	0.78%	0.53%	0.26% **	1.09%	0.96%	0.93%	0.72	0.55	0.28
12	3	0.32%	0.42%	-0.10%	2.00%	2.14%	1.69%	0.16	0.20	-0.06
12	6	0.26%	0.50%	-0.24% **	1.25%	1.62%	1.16%	0.21	0.31	-0.21
12	12	0.33%	0.46%	-0.13% **	0.74%	1.10%	0.88%	0.45	0.42	-0.15

*W = Winners, L = Losers. The corresponding t-statistics are enclosed in parentheses. Notations of *, **, and *** represents statistical significance at the 10%, 5%, and 1% levels, respectively.*

Table 4 shows further details of the returns of the ESG Momentum strategy across our tested portfolios. For shorter formation periods, we found that ESG Winners tend to produce better risk-adjusted returns, as observed with the higher Sharpe ratios, compared to ESG Losers. While for longer formation period, we often observe the ESG Losers will have higher Sharpe ratios, yet the contrast between ESG Winners is comparatively less stark when compared to shorter formation periods.

When it comes to volatility, the comprehensive nature of how ESG ratings are calculated means that we should expect ESG Losers should somewhat experience higher volatility as lower ESG ratings should mean higher unmanaged ESG risks. Yet, we only observe this phenomenon in longer formation periods where in the F12 portfolios, ESG Losers tends to have lower volatility, but we see less clear relationship when it comes to F3 and F6 portfolios.

4.2. Implementation of Price Momentum Tilt

Table 5: ESG Momentum Strategies Returns with Price Momentum Tilt

Formation Period	Holding Period	3			6			12		
		Returns	FF3 Alpha		Returns	FF3 Alpha		Returns	FF3 Alpha	
3	<i>W (ESG, Price)</i>	0.76%			1.28%			0.99%		
	<i>L (ESG, Price)</i>	0.53%			0.47%			1.07%		
	<i>W - L</i>	0.23%	0.32%	(0.93)	0.80% **	0.75%	(2.53)	-0.08%	-0.21%	(-0.69)
6	<i>W (ESG, Price)</i>	0.90%			0.83%			0.68%		
	<i>L (ESG, Price)</i>	0.55%			0.50%			0.52%		
	<i>W - L</i>	0.35%	0.29%	(0.72)	0.33%	0.24%	(0.91)	0.16%	0.10%	(0.56)
12	<i>W (ESG, Price)</i>	0.15%			-0.01%			0.14%		
	<i>L (ESG, Price)</i>	0.22%			0.37%			0.46%		
	<i>W - L</i>	-0.07%	-0.04%	(-0.1)	-0.38%	-0.35%	(-1.35)	-0.33% *	-0.33%	(-1.77)

*W = Winners, L = Losers. The corresponding t-statistics are enclosed in parentheses. Notations of *, **, and *** represents statistical significance at the 10%, 5%, and 1% levels, respectively.*

We further enhance our previous strategy by integrating a price momentum tilt, so called the ESG-Price Momentum strategy. This integration involves identifying ‘Winners’ that not only exhibit substantial upgrades in ESG ratings but have also demonstrate strong historical returns during our formation periods. By incorporating this additional criterion, our investment approach seeks to capitalize on companies that have both a positive ESG trajectory and a track record of favourable price performance. This involves categorizing stocks into two quintile groups of price momentum and ESG momentum, totalling to 25 groups for each of our 9 combination of formation and holding periods. We then buy stocks that are categorized as winners in both criteria (high past returns, high ESG momentum) and short the losers (low past returns, low ESG momentum).

Table 5 shows the results of this ESG-Price Momentum strategy. Interestingly, we still observe that the Loser groups manage to generate positive returns, despite comprising of companies with ESG rating downgrades and relatively low past returns. Compared to our ESG Momentum strategy, now we observe only one portfolio that yielded statistically significant positive return, the F3-H6 portfolio.

With the ESG Momentum strategy, the F3-H6 portfolio which yielded significant returns of 0.58%, the highest amongst our portfolios, now returns an additional 0.17% with the ESG-Price momentum strategy. This enhancement is mostly produced by buying stocks with the best performance in terms of ESG movements and past returns, which yielded 1.28% versus the 0.97% by solely using ESG momentum strategy. Equally important is that our FF3 alpha also appreciated by +0.36% to 0.93%.

4.3. Portfolio Construction using Individual ESG Components

Table 6: Monthly Returns of Momentum Strategy by ESG Components

Formation	Holding	Environmental			Social			Governance		
		W	L	W - L	W	L	W - L	W	L	W - L
3	3	0.77%	0.40%	0.37% **	0.60%	0.45%	0.15%	0.86%	0.60%	0.26%
3	6	1.07%	0.69%	0.38% *	0.79%	0.30%	0.50% ***	1.09%	0.52%	0.57% ***
3	12	1.19%	0.94%	0.24%	0.98%	0.63%	0.35% **	1.25%	0.69%	0.56% **
6	3	0.51%	0.75%	-0.24%	0.60%	0.51%	0.09%	0.56%	0.66%	-0.10%
6	6	0.78%	0.54%	0.24%	0.60%	0.55%	0.05%	0.64%	0.86%	-0.21%
6	12	0.84%	0.59%	0.25% **	0.65%	0.61%	0.04%	0.72%	0.85%	-0.13%
12	3	0.29%	0.41%	-0.11%	0.47%	0.42%	0.06%	0.42%	0.32%	0.10%
12	6	0.26%	0.46%	-0.21% ***	0.59%	0.49%	0.10%	0.44%	0.41%	0.03%
12	12	0.34%	0.49%	-0.16% ***	0.57%	0.51%	0.06%	0.45%	0.37%	0.08%

W = Winners, L = Losers. Notations of *, **, and *** represents statistical significance at the 10%, 5%, and 1% levels, respectively.

The ESG ratings that we have used so far comprises weighted ratings for environmental, social and governance ratings according to the risk exposure for the covered companies. In this next section, using the same approach that with the ESG momentum strategy, we used the respective Environmental (E), Social (S) and Governance (G) ratings to analyse the consistency and robustness of our results that we have obtained with previous analyses.

Table 6 shows the results of using individual E, S and G ratings in constructing the Momentum strategy. Consistent with our previous strategies, the F3-H6 strategy remains the sole portfolio that delivers positive and statistically significant returns for when replicating our strategy using individual E, S and G ratings. The E Momentum strategy generated the closest results with our ESG Momentum in terms of significance of returns across the nine portfolios. Specifically, we see that the F12 portfolios yielded significantly negative returns when utilizing environmental ratings, but not when using social or governance scores.

4.4. Cumulative Performance and Benchmarking

Figure 2: Cumulative Performance of ESG Momentum Strategy versus FTSE 100 Index (blue)



Rows: F3, F6, F12. Columns: -H3, -H6, -H12

Figure 2 shows the cumulative returns of ESG Momentum compared to two benchmark indexes: the FTSE100 and S&P500 which captures how our strategy performs within our sample period.

Out of 9, two portfolios managed to outperform the FTSE100 Index, the F3-H3 (top right) and F3-H6 (top middle) portfolios. The F3-H6 strategy, which is statistically significant, had the best overall performance, returning 107.4% versus the benchmark 39.7%.

5. LIMITATIONS

Our study's limitation concerns the methodology underlying ESG ratings. Because ESG considerations in investments were recently incorporated, corporate ESG ratings don't cover the entire securities universe. This incomplete coverage hinders a thorough analysis of ESG momentum strategy performance and introduces selection bias.

Similar to other agencies, Sustainalytics periodically revises their rating methods to include advances in corporate ESG reporting, regulations, and stakeholder expectations. While this enhances ESG risk assessment, it creates inconsistencies for cross-sectional ESG momentum comparisons, especially over longer periods. Evolving methodologies might not align with updated criteria, complicating year-to-year comparisons.

The variety of methodologies ESG rating agencies employ is a significant limitation in assessing ESG momentum strategy performance. The lack of standardization, changing ESG

regulations, and diverse agency approaches lead to discrepancies in evaluating companies' ESG practices. Consequently, ESG ratings from different agencies can never be truly identical.

6. CONCLUSION

Using the stock universe of British companies covered by Sustainalytics ESG Ratings, our comprehensive analysis of the ESG Momentum strategy have shown that a strategy with shorter formation periods tends to deliver more consistent, statistically significant and positive excess returns compared to strategies with longer formation periods. In fact, our ESG Momentum strategies with formation periods of 12 months tend to produce negative excess returns – even after stripping growth and value factors. This aligns with the findings of Sverner (2022) which acknowledges the existence of an ‘ESG Risk Premium’ for lower ESG-rated companies. We could infer that over long formation and holding periods, the market would have priced in said risk premium, resulting in ESG Losers to have higher returns compared to Winners. Whereas in shorter observation periods, market adjusts relatively slowly and results in higher risk-adjusted returns for ESG Winners, making the ESG Momentum a viable strategy.

An additional finding is that ESG Losers consistently generated positive average monthly returns across all our portfolios. This observation indicates that companies with increasing ESG risks could still yield positive stock returns despite having experienced ESG rating downgrades. Therefore, ESG Momentum strategies that shorts ESG Losers does not benefit from better risk-adjusted returns as we see no consistent improvements in Sharpe ratios. One reason for this is due to the increase in ESG risk could possibly enhance the beta of companies and therefore increases the expected return premium for investors owing these stocks. This find captures the core finding of Ciciretti et al. (2023) which shows that “firms that score high on ESG indicators exhibit lower expected returns” and could be attributed to investors’ preferences for companies with improving ESG scores.

In the context of shorter formation periods, the ESG Momentum strategy displays the potential to outperform the local benchmark index. However, this advantage holds true only for shorter holding periods as well. This observation is grounded in the fact that ESG Winners consistently outpace the performance of ESG Losers, despite the latter demonstrating favourable risk-adjusted returns within briefer holding periods. Although Winners tend to exhibit heightened volatility, this is offset by their ability to deliver superior returns per unit of volatility, ultimately leading to higher Sharpe ratios compared to Losers.

Additionally, our analysis underscores that the integration of a price momentum tilt further enhances the effectiveness of our ESG Momentum strategy. This tilt enables us to selectively target companies poised for stock price appreciation in tandem with improved ESG ratings, rather than solely relying on ESG rating upgrades. As a result, this strategic adjustment contributes to notable improvements in the returns generated by the ESG Momentum strategy, enhancing them by approximately 0.19% across our diverse set of nine portfolios.

In summation, our analysis underscores the potential of the ESG Momentum strategy, showcasing its effectiveness in generating positive excess returns within shorter formation and

holding periods. The integration of price momentum tilt further enhances the strategy's performance, enabling a nuanced approach to identifying companies poised for stock price growth alongside ESG rating improvements. This holistic strategy adjustment reflects a valuable commitment to achieving sustainable financial outcomes while considering ESG factors.

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8. APPENDIX

Loading Price Data

```
dprice_data <- fread("compustat1.csv")

dprice_data <- subset(dprice_data, !is.na(isin) & isin != "")
length(unique(dprice_data$isin))
tail(dprice_data)

## Filtering first and last day of each month
dprice_data <- dprice_data %>%
  mutate(monthYear = format(datadate, "%Y-%m")) %>%
  ungroup() %>%
  as.data.table()

dprice_data <- dprice_data %>%
  group_by(isin, monthYear) %>%
  mutate(first_day_price = first(prccd), last_day_price = last(prccd), monthly_return = (last_day_price -
    first_day_price)/first_day_price) %>%
  select(-c(first_day_price, last_day_price)) %>%
  as.data.table()

dprice_data$monthly_return[is.infinite(dprice_data$monthly_return)] <- NA
dprice_data$monthly_return[(dprice_data$price == 0)] <- NA
dprice_data$datadate <- as.Date(dprice_data$datadate, format = "%Y-%m-%d", origin = "1970-01-01")

output_file <- "dprice_data.RData"

# Save the dprice_data dataframe in R's native file format
saveRDS(dprice_data, file = output_file)
```

Loading ESG Data

```
# Using the fread() function to load the dataset faster

dprice_data <- readRDS("dprice_data.RData")

esg_data1 <- fread("sustain1.csv")
esg_data1 <- subset(esg_data1, Type_Listing == "Public" & !is.na(ISIN) & ISIN !=
  "" & ISIN != "No data")

common_isin_values <- intersect(dprice_data$isin, esg_data1$ISIN)

dprice_data <- dprice_data %>%
  filter(isin %in% common_isin_values) %>%
  as.data.table()

esg_data1 <- esg_data1 %>%
  mutate(Date = as.Date(Date, format = "%d/%m/%Y")) %>%
  mutate(Date = format(Date, "%Y-%m-%d")) %>%
  arrange(Date) %>%
  as.data.table()

esg_data1 <- esg_data1 %>%
  mutate(monthYear = format(as.Date(Date), "%Y-%m")) %>%
  as.data.table()

esg_data1 <- esg_data1 %>%
  rename(datadate = Date) %>%
  as.data.table()

esg_data1$datadate <- as.Date(esg_data1$datadate, format = "%Y-%m-%d", origin = "1970-01-01")

esg_data1 <- esg_data1 %>%
  group_by(monthYear, ISIN) %>%
  filter(datadate == max(datadate)) %>%
  ungroup() %>%
  as.data.table()
```

```

# Create a sequence of all months from 01 to 12
months <- sprintf("%02d", 1:12)

# Create a sequence of years from 2007 to 2020
years <- 2009:2019

# Expand the sequences to match all possible combinations of years and months
all_dates <- expand.grid(year = years, month = months)

# Create the 'monthYear' column by combining the year and month
all_dates$monthYear <- paste(all_dates$year, all_dates$month, sep = "-")

# Convert the data.frame to a data.table if needed
all_dates <- all_dates %>%
  arrange(monthYear) %>%
  as.data.table()

# Get unique isin and monthYear values
unique_isin <- unique(common_isin_values)
unique_monthYear <- unique(all_dates$monthYear)

# Create a data frame with all combinations of isin and monthYear
esg_data2 <- expand.grid(monthYear = unique_monthYear, ISIN = unique_isin) %>%
  as.data.table()

esg_data2 <- merge(esg_data2, esg_data1[, .(ISIN, monthYear, total_esg_score, environment_score,
  social_score, governance_score)], by = c("ISIN", "monthYear"), all.x = TRUE)

is_duplicate <- duplicated(esg_data2[, c("ISIN", "monthYear")])

# Get the duplicate rows
duplicate_rows <- esg_data2[is_duplicate, ]

# Print the duplicate rows print(duplicate_rows)

esg_data2 <- esg_data2 %>%
  distinct(ISIN, monthYear, .keep_all = TRUE) %>%
  as.data.table()

esg_data <- esg_data2 %>%
  arrange(monthYear) %>%
  group_by(ISIN) %>%
  mutate(esg_momentum = total_esg_score - lag(total_esg_score, 1)) %>%
  mutate(env_momentum = environment_score - lag(environment_score, 1)) %>%
  mutate(gov_momentum = governance_score - lag(governance_score, 1)) %>%
  mutate(soc_momentum = social_score - lag(social_score, 1)) %>%
  ungroup() %>%
  as.data.table()

esg_data <- esg_data %>%
  rename(isin = ISIN) %>%
  as.data.table()

rm(all_dates, duplicate_rows, esg_data1, esg_data2)
rm(is_duplicate, months, common_isin_values, years)

```

Loading Fama-French Data

```

fama5_data <- fread("fama5factors.csv")

fama5_data <- fama5_data %>%
  arrange(dateff) %>%
  as.data.table()

fama5_data <- fama5_data %>%
  mutate(monthYear = format(as.Date(dateff), "%Y-%m")) %>%
  as.data.table()

fama5_data$dateff <- as.Date(fama5_data$dateff, format = "%Y-%m-%d", origin = "1970-01-01")

```

Merging Datasets

```
merged_data <- merge(esg_data, dprice_data, by = c("isin", "monthYear")) %>%
  as.data.table()

temp1 <- dprice_data %>%
  group_by(isin, monthYear) %>%
  summarise(monthly_return) %>%
  unique() %>%
  as.data.table()

merged_data <- merge(esg_data, temp1[, .(isin, monthYear, monthly_return)], by = c("isin",
  "monthYear"), all.x = TRUE)

#rm(combinations, data1, data2, data3, data4, data5, gooduniverse, gooduniverse_temp, merged1)
```

Loading Benchmark Data

```
ftse_data <- fread("ftse100.csv")
ftse_data <- ftse_data %>%
  mutate(monthYear = format(as.Date(date), "%Y-%m"), monthYear = lead(monthYear)) %>%
  as.data.table()
ftse_data$date <- as.Date(ftse_data$date, format = "%Y-%m-%d", origin = "1970-01-01") %>%
  as.data.table()

sp_data <- fread("sp500.csv")
sp_data <- sp_data %>%
  mutate(monthYear = format(as.Date(date), "%Y-%m"), monthYear = lead(monthYear)) %>%
  as.data.table()
sp_data$date <- as.Date(sp_data$date, format = "%Y-%m-%d", origin = "1970-01-01") %>%
  as.data.table()
```

ESG Momentum Dataset Construction

```
mom <- merged_data %>%
  select(-c(esg_momentum, env_momentum, soc_momentum, gov_momentum)) %>%
  group_by(isin) %>%
  mutate(esg_mom3 = rollapply(total_esg_score, 4, function(x) (x[4] - x[1]), fill = NA,
    align = "right"), env_mom3 = rollapply(environment_score, 4, function(x) (x[4] -
    x[1]), fill = NA, align = "right"), soc_mom3 = rollapply(social_score, 4,
    function(x) (x[4] - x[1]), fill = NA, align = "right"), gov_mom3 = rollapply(governance_score,
    4, function(x) (x[4] - x[1]), fill = NA, align = "right"), p_mom3 = rollapply(monthly_return,
    3, function(x) prod(x + 1) - 1, fill = NA, align = "right")) %>%
  mutate(esg_mom6 = rollapply(total_esg_score, 7, function(x) (x[7] - x[1]), fill = NA,
    align = "right"), env_mom6 = rollapply(environment_score, 7, function(x) (x[7] -
    x[1]), fill = NA, align = "right"), soc_mom6 = rollapply(social_score, 7,
    function(x) (x[7] - x[1]), fill = NA, align = "right"), gov_mom6 = rollapply(governance_score,
    7, function(x) (x[7] - x[1]), fill = NA, align = "right"), p_mom6 = rollapply(monthly_return,
    6, function(x) prod(x + 1) - 1, fill = NA, align = "right")) %>%
  mutate(esg_mom12 = rollapply(total_esg_score, 13, function(x) (x[13] - x[1]),
    fill = NA, align = "right"), env_mom12 = rollapply(environment_score, 13,
    function(x) (x[13] - x[1]), fill = NA, align = "right"), soc_mom12 = rollapply(social_score,
    13, function(x) (x[13] - x[1]), fill = NA, align = "right"), gov_mom12 = rollapply(governance_
score,
    13, function(x) (x[13] - x[1]), fill = NA, align = "right"), p_mom12 = rollapply(monthly_retur
n,
    12, function(x) prod(x + 1) - 1, fill = NA, align = "right")) %>%
  mutate(return_temp = rollapply(monthly_return, 3, function(x) prod(1 + x) - 1,
    fill = NA, align = "right"), return_3mo = return_temp/3) %>%
  select(-return_temp) %>%
  mutate(return_temp = rollapply(monthly_return, 6, function(x) prod(1 + x) - 1,
    fill = NA, align = "right"), return_6mo = return_temp/6) %>%
  select(-return_temp) %>%
  mutate(return_temp = rollapply(monthly_return, 12, function(x) prod(1 + x) -
    1, fill = NA, align = "right"), return_12mo = return_temp/12) %>%
  select(-return_temp) %>%
  ungroup() %>%
  as.data.table()

mom <- mom %>%
  group_by(monthYear) %>%
  mutate(across(c(p_mom3, esg_mom3, env_mom3, soc_mom3, gov_mom3), ~if_else(lag(.x,
    3) != 0 & !is.na(lag(.x, 3)), ntile(lag(.x, 3), 5), NA_integer_), .names = "{.col}_q"),
    across(c(p_mom6, esg_mom6, env_mom6, soc_mom6, gov_mom6), ~if_else(lag(.x,
    6) != 0 & !is.na(lag(.x, 6)), ntile(lag(.x, 6), 5), NA_integer_), .names = "{.col}_q"),
```

```

    across(c(p_mom12, esg_mom12, env_mom12, soc_mom12, gov_mom12), ~if_else(lag(.x,
    12) != 0 & !is.na(lag(.x, 12)), ntile(lag(.x, 12), 5), NA_integer_),
    .names = "{.col}_q") %>%
ungroup() %>%
as.data.table()

```

Portfolio Construction

```

ret5 <- mom %>%
  filter(esg_mom3_q == 5) %>%
  group_by(monthYear) %>%
  summarise(return = mean(return_12mo, na.rm = TRUE)) %>%
  as.data.table()

ret4 <- mom %>%
  filter(esg_mom3_q == 4) %>%
  group_by(monthYear) %>%
  summarise(return = mean(return_12mo, na.rm = TRUE)) %>%
  as.data.table()

ret3 <- mom %>%
  filter(esg_mom3_q == 3) %>%
  group_by(monthYear) %>%
  summarise(return = mean(return_12mo, na.rm = TRUE)) %>%
  as.data.table()

ret2 <- mom %>%
  filter(esg_mom3_q == 2) %>%
  group_by(monthYear) %>%
  summarise(return = mean(return_12mo, na.rm = TRUE)) %>%
  as.data.table()

ret1 <- mom %>%
  filter(esg_mom3_q == 1) %>%
  group_by(monthYear) %>%
  summarise(return = mean(return_12mo, na.rm = TRUE)) %>%
  as.data.table()

tables_list <- list(ret5, ret4, ret3, ret2, ret1)

# Perform a full join for all tables based on the monthYear column
mom_ret <- reduce(tables_list, full_join, by = "monthYear") %>%
  as.data.table()

mom_ret_F3H12 <- mom_ret %>%
  rename_at(vars(2:6), ~c("top20", "4", "3", "2", "bottom20")) %>%
  as.data.table()

rm(ret1, ret2, ret3, ret4, ret5, tables_list)

```

Saving ESG Momentum Portfolio Data

```

saveRDS(mom_ret_F12H6, "mom3_ret_F12H6.RData")
saveRDS(mom_ret_F12H3, "mom3_ret_F12H3.RData")
saveRDS(mom_ret_F12H12, "mom3_ret_F12H12.RData")

saveRDS(mom_ret_F6H6, "mom3_ret_F6H6.RData")
saveRDS(mom_ret_F6H3, "mom3_ret_F6H3.RData")
saveRDS(mom_ret_F6H12, "mom3_ret_F6H12.RData")

saveRDS(mom_ret_F3H6, "mom3_ret_F3H6.RData")
saveRDS(mom_ret_F3H3, "mom3_ret_F3H3.RData")
saveRDS(mom_ret_F3H12, "mom3_ret_F3H12.RData")

```

Loading ESG Momentum Portfolios

```

mom_ret_F12H6 <- readRDS("mom3_ret_F12H6.RData")
mom_ret_F12H3 <- readRDS("mom3_ret_F12H3.RData")
mom_ret_F12H12 <- readRDS("mom3_ret_F12H12.RData")

mom_ret_F6H6 <- readRDS("mom3_ret_F6H6.RData")
mom_ret_F6H3 <- readRDS("mom3_ret_F6H3.RData")

```

```

mom_ret_F6H12 <- readRDS("mom3_ret_F6H12.RData")

mom_ret_F3H6 <- readRDS("mom3_ret_F3H6.RData")
mom_ret_F3H3 <- readRDS("mom3_ret_F3H3.RData")
mom_ret_F3H12 <- readRDS("mom3_ret_F3H12.RData")

replace_na_with_zero <- function(x) {
  ifelse(is.na(x), 0, x)
}

table_list <- list(
  mom_ret_F3H3,
  mom_ret_F3H6,
  mom_ret_F3H12,

  mom_ret_F6H3,
  mom_ret_F6H6,
  mom_ret_F6H12,

  mom_ret_F12H3,
  mom_ret_F12H6,
  mom_ret_F12H12
)

```

Statistical Summary across Portfolios

```

compile <- function(table_list) {
  result_list <- lapply(table_list, function(data) {
    # Calculate mean and convert to data.table for each table in the list
    data <- merge(data, fama5_data[, .(monthYear, rf)], by = "monthYear", all.x = TRUE)

    temp <- data %>%
      select(-monthYear) %>%
      mutate(across(everything(), ~. - rf)) %>%
      mutate(topbottom = top20 - bottom20) %>%
      select(-rf) %>%
      summarise(across(everything(), mean, na.rm = TRUE)) %>%
      as.data.table()

    return(temp)
  })

  # Combine the rows of all data.tables in the result_list using rbindlist
  result <- rbindlist(result_list)

  return(result)
}

# Assuming you have the List of tables 'table_list' available Call the function
# and pass the List of tables as an argument
result <- compile(table_list)

```

Fama-French 3 Factor Regression

```

# Function to compile regression results
compile_regression_results <- function(data) {
  data[, names(data) := lapply(.SD, function(x) replace(x, x == 0, NA)), .SDcols = names(data)]

  data <- merge(data, fama5_data[, .(monthYear, mktrf, rf, smb, hml)], by = "monthYear",
    all.x = TRUE)

  data <- data %>%
    mutate(topminusbottom = top20 - bottom20 - rf) %>%
    as.data.table()

  model <- lm(topminusbottom ~ mktrf + smb + hml, data = data)

  coefficients <- summary(model)$coefficients[, "Estimate"]
  std_errors <- summary(model)$coefficients[, "Std. Error"]
  adjusted_r_squared <- summary(model)$adj.r.squared

```

```

# Combine coefficients, standard errors, and adjusted R-squared into a
# data.table
regression_table <- data.table(Coefficient = coefficients, `Standard Error` = std_errors,
                              `Adjusted R-squared` = adjusted_r_squared)

return(regression_table)
}

# Assuming you have the list of tables 'table_list' available Create an empty
# list to store the results
regression_results_list <- list()

# Loop through each table in the 'table_list' and apply the function to compile
# regression results
for (data in table_list) {
  regression_result <- compile_regression_results(data)
  regression_results_list <- c(regression_results_list, list(regression_result))
}

# Combine the results for all tables into a single data.table using rbindlist
final_regression_results <- rbindlist(regression_results_list)

```

T-test Regression Results

```

# Function to compile regression results
compile_regression_results <- function(data) {
  data[, names(data) := lapply(.SD, function(x) replace(x, x == 0, NA)), .SDcols = names(data)]

  data <- merge(data, fama5_data[, .(monthYear, mktrf, rf, smb, hml)], by = "monthYear",
               all.x = TRUE)

  data <- data %>%
    mutate(topminusbottom = top20 - bottom20 - rf) %>%
    as.data.table()

  model <- lm(topminusbottom ~ mktrf, data = data)

  coefficients <- summary(model)$coefficients[, "Estimate"]
  std_errors <- summary(model)$coefficients[, "Std. Error"]
  adjusted_r_squared <- summary(model)$adj.r.squared

  # Combine coefficients, standard errors, and adjusted R-squared into a
  # data.table
  regression_table <- data.table(Coefficient = coefficients, `Standard Error` = std_errors,
                              `Adjusted R-squared` = adjusted_r_squared)

  return(regression_table)
}

# Assuming you have the list of tables 'table_list' available Create an empty
# list to store the results
regression_results_list <- list()

# Loop through each table in the 'table_list' and apply the function to compile
# regression results
for (data in table_list) {
  regression_result <- compile_regression_results(data)
  regression_results_list <- c(regression_results_list, list(regression_result))
}

# Combine the results for all tables into a single data.table using rbindlist
final_regression_results <- rbindlist(regression_results_list)

reg_fama <- mom_ret_F12H12 %>%
  as.data.table()

reg_fama[, names(reg_fama) := lapply(.SD, function(x) replace(x, x == 0, NA)),
         .SDcols = names(reg_fama)]

reg_fama <- merge(reg_fama, fama5_data[, .(monthYear, mktrf, rf, smb, hml)], by = "monthYear",
               all.x = TRUE)

```

```

reg_fama <- reg_fama %>%
  mutate(topminusbottom = top20 - bottom20 - rf) %>%
  as.data.table()

model <- lm(topminusbottom ~ mktrf + smb + hml, data = reg_fama)
summary(model)

reg_fama <- mom_ret_F12H12 %>%
  as.data.table()

reg_fama[, names(reg_fama) := lapply(.SD, function(x) replace(x, x == 0, NA)),
  .SDcols = names(reg_fama)]

reg_fama <- merge(reg_fama, fama5_data[, .(monthYear, mktrf, rf, smb, hml)], by = "monthYear",
  all.x = TRUE)

reg_fama <- reg_fama %>%
  mutate(topminusbottom = top20 - bottom20 - rf) %>%
  as.data.table()

# Perform one-sample t-test on the 'topminusbottom' variable
coefficient_test <- t.test(reg_fama$topminusbottom, mu = 0)

# Check the p-value to determine significance
p_value <- coefficient_test$p.value

# Convert p-value to significance codes
signif_codes <- symnum(p_value, corr = FALSE, na = FALSE, cutpoints = c(0, 0.001,
  0.01, 0.05, 0.1, 1), symbols = c("***", "**", "*", ".", " "))

if (p_value < 0.05) {
  # The 'topminusbottom' variable is significantly different from zero
  print(paste("The 'topminusbottom' variable is significantly different from zero. Significance code
:",
    signif_codes))
} else {
  # The 'topminusbottom' variable is not significantly different from zero
  print(paste("The 'topminusbottom' variable is not significantly different from zero. Significance
code:",
    signif_codes))
}

```

Portfolio Performance and Benchmarking

```

# Replace NA with zero for mom_ret_F3H6
mom_ret_F3H6[] <- lapply(mom_ret_F3H6, replace_na_with_zero)

# Replace NA with zero for mom_ret_F3H3
mom_ret_F3H3[] <- lapply(mom_ret_F3H3, replace_na_with_zero)

# Replace NA with zero for mom_ret_F3H12
mom_ret_F3H12[] <- lapply(mom_ret_F3H12, replace_na_with_zero)

for (table in table_list) {
  table[] <- lapply(table, replace_na_with_zero)
}

generate_cumulative_plot <- function(data) {
  temp <- data %>%
    select(monthYear, top20, bottom20) %>%
    mutate(momret = top20 - bottom20) %>%
    select(-c(top20, bottom20)) %>%
    as.data.table()

  temp <- merge(temp, sp_data[, .(monthYear, change)], by = "monthYear", all.x = TRUE)

  # Convert monthYear to a proper date format
  temp$monthYear <- as.Date(paste0(temp$monthYear, "-25"))

  # Calculate the cumulative performance for each variable
  temp <- temp %>%
    arrange(monthYear) %>%

```

```

mutate(cumulative_change = cumprod(1 + change) * 100, cumulative_momret = cumprod(1 +
momret) * 100) %>%
as.data.table()

# Plot the cumulative performance
plot <- ggplot(temp, aes(x = monthYear)) + geom_line(aes(y = cumulative_change,
color = "FTSE 100")) + geom_line(aes(y = cumulative_momret, color = "ESG Momentum")) +
labs(x = NULL, y = NULL, color = "Variable") + theme_minimal() + guides(color = FALSE) +
coord_cartesian(ylim = c(70, 270)) # Set the y-axis limits to [70, 210] #210 for ftse

return(plot)
}

cum_F3H12 <- generate_cumulative_plot(mom_ret_F3H12)
cum_F3H3 <- generate_cumulative_plot(mom_ret_F3H3)
cum_F3H6 <- generate_cumulative_plot(mom_ret_F3H6)
cum_F6H12 <- generate_cumulative_plot(mom_ret_F6H12)
cum_F6H3 <- generate_cumulative_plot(mom_ret_F6H3)
cum_F6H6 <- generate_cumulative_plot(mom_ret_F6H6)
cum_F12H12 <- generate_cumulative_plot(mom_ret_F12H12)
cum_F12H3 <- generate_cumulative_plot(mom_ret_F12H3)
cum_F12H6 <- generate_cumulative_plot(mom_ret_F12H6)

# Combine the plots into a 3x3 grid

grid <- grid.arrange(cum_F3H3, cum_F3H6, cum_F3H12, cum_F6H3, cum_F6H6, cum_F6H12,
cum_F12H3, cum_F12H6, cum_F12H12, ncol = 3)

rm(cum_F3H3, cum_F3H6, cum_F3H12, cum_F6H3, cum_F6H6, cum_F6H12, cum_F12H3, cum_F12H6,
cum_F12H12)

plot(grid, xlim = c(80, 210), ylim = c(80, 210))

# ggsave('sp.png', plot = grid, width = 12, height = 8, dpi = 300)

```

FF3 Regression Data

```

datatemp <- mom_ret_F12H12

results_a <- data.frame()
results_b <- data.frame()
results_c <- data.frame()
results_d <- data.frame()
results_tval_a <- data.frame()
results_tval_b <- data.frame()
results_adj_r2 <- data.frame()

datatemp <- merge(datatemp, fama5_data[, .(monthYear, mktrf, rf, smb, hml)], by = "monthYear",
all.x = TRUE) %>%
as.data.table()

for (col in c(2:6)) {
model <- lm((datatemp[[col]] - datatemp$rf) ~ datatemp$mktrf + datatemp$smb +
datatemp$hml) # + datatemp$smb + datatemp$hml
results_a <- rbind(results_a, coef(model)[1]) #alpha
results_b <- rbind(results_b, coef(model)[2]) #beta
results_c <- rbind(results_c, coef(model)[3]) #smb
results_d <- rbind(results_d, coef(model)[4]) #hml
results_tval_a <- rbind(results_tval_a, coef(summary(model))[1, "t value"])
results_tval_b <- rbind(results_tval_b, coef(summary(model))[2, "t value"])
results_adj_r2 <- rbind(results_adj_r2, data.frame(adj_r2 = summary(model)$adj.r.squared))
}

mean_ret <- matrix(colMeans(datatemp[, c(2:6)] - datatemp$rf, na.rm = TRUE)) #value-weighted
a <- matrix(results_a[, 1]) #value-weighted
b <- matrix(results_b[, 1]) #value-weighted
t_alphas <- matrix(results_tval_a[, 1]) #value-weighted
t_betas <- matrix(results_tval_b[, 1]) #value-weighted
adj_r2 <- matrix(results_adj_r2[, 1]) #value-weighted

```

```

alphas <- data.frame(combo = paste(round(results_a[, 1], 3), " (", round(results_tval_a[,
  1], 3), ")", sep = ""))
alphas <- matrix(alphas[, 1])
betas <- data.frame(combo = paste(round(results_b[, 1], 3), " (", round(results_tval_b[,
  1], 3), ")", sep = ""))
betas <- matrix(betas[, 1])

alphas <- xtable(alphas)
betas <- xtable(betas)
mean_ret <- xtable(round(mean_ret, 3))
adj_r2 <- xtable(round(adj_r2, 3))
colnames(alphas) <- c("alphas")
rownames(alphas) <- c("Top20", "Q4", "Q3", "Q2", "Bottom20")
colnames(betas) <- c("betas")
rownames(betas) <- c("Top20", "Q4", "Q3", "Q2", "Bottom20")
colnames(mean_ret) <- c("meanreturn")
rownames(mean_ret) <- c("Top20", "Q4", "Q3", "Q2", "Bottom20")
colnames(adj_r2) <- c("adjr2")
rownames(adj_r2) <- c("Top20", "Q4", "Q3", "Q2", "Bottom20")

kable(mean_ret, align = "c", caption = "Portfolio Average Monthly Excess Returns")
kable(betas, align = "c", caption = "Portfolio Betas (t-values)")
kable(alphas, align = "c", caption = "Portfolio Alphas (t-values)")
kable(adj_r2, align = "c", caption = "Adjusted R-Squared")

```

Expected vs Actual Returns

```

rm(capm_plot)
capm_plot <- results_b*mean(datatemp$mktrf) + results_c*mean(datatemp$smb) + results_d*mean(datatemp$sh
ml)
capm_plot$act_ret <- colMeans(datatemp[, 2:6] - datatemp$rf, na.rm = TRUE)
colnames(capm_plot)[1] <- "pred_ret"

plot(capm_plot$pred_ret, capm_plot$act_ret,
     pch = 16, col = "blue",
     #main = "Predicted vs Actual Excess Returns",
     xlab = "Predicted Excess Return",
     ylab = "Actual Excess Return",
     xlim = c(min(capm_plot$pred_ret) - 0.0001, max(capm_plot$pred_ret) + 0.0001),
     ylim = c(min(capm_plot$pred_ret) - 0.001, max(capm_plot$act_ret) + 0.001))
abline(a = 0, b = 1, col = "red")

points_labels <- c("Top", "ESG4", "ESG3", "ESG2", "Bottom")
points_x <- c(capm_plot$pred_ret[1], capm_plot$pred_ret[2], capm_plot$pred_ret[3], capm_plot$pred_ret[
4], capm_plot$pred_ret[5])
points_y <- c(capm_plot$act_ret[1], capm_plot$act_ret[2], capm_plot$act_ret[3], capm_plot$act_ret[4],
capm_plot$act_ret[5])

# Adding a line that joins the points in order of labels
order_indices <- order(match(points_labels, unique(points_labels))) # Get the order of indices based
on labels
lines(points_x[order_indices], points_y[order_indices], col = "black")

# Adding Labels
for (i in 1:length(points_labels)) {
  text(points_x[i], points_y[i], labels = points_labels[i], pos = 3)
}

```